

## Chapter No 3

### Matrices

#### Matrix

A matrix is a function that consist of an ordered rectangular array of the numbers.

- vertical row called column.
- Horizontal row called row.

#### Types of Matrix

##### Row Matrix

A matrix which have only one row.

i.e  $A = [1 \ 2 \ 3]$

##### Column Matrix

A matrix which have only one column.

i.e  $A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

##### Rectangular Matrix

A matrix in which ~~has~~ the number of rows is not equal to the number of columns

i.e  $A = \begin{bmatrix} 2 & 3 & 1 \\ -1 & 0 & 4 \end{bmatrix}$

## Square Matrix

The matrix which have the same number of rows and columns.

i.e  $A = \begin{bmatrix} 0 \end{bmatrix}$        $B = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix}$

## Diagonal Matrix

A square matrix  $A$  is called a diagonal matrix if at least any one of the entries of its diagonal is not zero and non-diagonal entries are zero.

i.e

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

## Scalar Matrix

A diagonal matrix is called a scalar matrix if all the diagonal entries are same and non-zero

i.e  $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

$$B = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$



## Unit Matrix or Identity Matrix

A diagonal matrix is called Identity (Unit matrix) if all diagonal entries are 1. It is denoted by I.

i.e

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

## Null or Zero Matrix

A matrix is called a null or zero matrix if each of its entries is 0.

i.e

$$A = \begin{bmatrix} 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

## Equal Matrix

\* } A matrix in which the number of rows is equal to the columns. \* }

Two matrix of the same order are said to be equal if their crossponding entries are equal

ie

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

## Symmetric Matrix

A Square matrix for which  $A^t = A$  is said to be a symmetric.

$$\text{i.e. } A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & -3 \\ 3 & -3 & 6 \end{bmatrix}$$

$$A^t = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & -3 \\ 3 & -3 & 6 \end{bmatrix}$$

$$A^t = A \quad \text{Prove it}$$

## Skew-Symmetric Matrix

A square matrix  $A$  for which  $A^t = -A$  is said to be a skew-symmetric Matrix

i.e.

$$A = \begin{bmatrix} 0 & -2 & 3 \\ 2 & 0 & 4 \\ -3 & -4 & 0 \end{bmatrix}$$

$$A^t = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & -4 \\ 3 & 4 & 0 \end{bmatrix}$$

$$= - \begin{bmatrix} 0 & -2 & 3 \\ 2 & 0 & 4 \\ -3 & -4 & 0 \end{bmatrix}$$

$$A^t = -A \quad \text{Prove it}$$



## Hermitian Matrix

A Square Matrix  $A$  ~~is~~ such that  $(\overline{A})^t = A$  is called Hermitian matrix.

i.e

$$A = \begin{bmatrix} 1 & 1-i & 3 \\ 1+i & 3 & i \\ 3 & -i & 0 \end{bmatrix}$$

$$\overline{A} = \begin{bmatrix} 1 & 1+i & 3 \\ 1-i & 3 & -i \\ 3 & +i & 0 \end{bmatrix}$$

$$(\overline{A})^t = \begin{bmatrix} 1 & 1-i & 3 \\ 1+i & 3 & i \\ 3 & -i & 0 \end{bmatrix}$$

$$(\overline{A})^t = A \quad \text{Prove it}$$

## Skew-Hermitian Matrix

A square matrix such that  $(\overline{A})^t = -A$  is called skew Hermitian matrix.

i.e

$$A = \begin{bmatrix} i & 1-i & a \\ -1-i & 3i & i \\ -a & i & 0 \end{bmatrix}$$

### Exercise No 3.1

Question no 1.

Let

$$A = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 3 \end{bmatrix}$$

Find

i)  $A+B$

$$A+B = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} + \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2-1 & -3+3 & -5+5 \\ -1+1 & 4-3 & 5-5 \\ 1-1 & -3+3 & -4+3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$A-B$

$$A-B = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} - \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 3 \end{bmatrix}$$



$$= \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} + \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2+1 & -3-3 & -5-5 \\ -1-1 & 4+3 & 5+5 \\ 1+1 & -3-3 & -4-3 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & -6 & -10 \\ -2 & 7 & 10 \\ 2 & -6 & -7 \end{bmatrix}$$

**2A + 3B**

$$2A + 3B = 2 \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} + 3 \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & -6 & -10 \\ -2 & 8 & 10 \\ 2 & -6 & -8 \end{bmatrix} + \begin{bmatrix} -3 & 9 & 15 \\ 3 & -9 & -15 \\ -3 & 9 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 4-3 & -6+9 & -10+15 \\ -2+3 & 8-9 & 10-15 \\ 2-3 & -6+9 & -8+9 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 3 & 5 \\ 1 & -1 & -5 \\ -1 & 3 & 1 \end{bmatrix}$$

3A - 5B

$$3A - 5B = 3 \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} - 5 \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 6 & -9 & -15 \\ -3 & 12 & 15 \\ 3 & -9 & -12 \end{bmatrix} - \begin{bmatrix} -5 & 15 & 25 \\ 5 & -15 & -25 \\ -5 & 15 & 15 \end{bmatrix}$$

$$= \begin{bmatrix} 6+5 & -9-15 & -15-25 \\ -3-5 & 12+15 & 15+25 \\ 3+5 & -9-15 & -12+15 \end{bmatrix}$$

$$= \begin{bmatrix} 11 & -24 & -40 \\ -8 & 27 & 40 \\ 8 & -24 & -27 \end{bmatrix}$$

AB

$$AB = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -2-3+5 & 6+8-15 & 10+15-15 \\ 1+4-5 & -3-12+15 & -5-20+15 \\ -1-3+5 & 3+9-12 & 5+15+12 \end{bmatrix}$$



$$= \begin{bmatrix} 0 & 0 & 10 \\ 0 & 0 & 10 \\ 0 & 0 & 8 \end{bmatrix}$$

BA

$$BA = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 3 \end{bmatrix} \cdot \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} -2+3+5 & 3+12-15 & 5+15-20 \\ 2+3-5 & -3-12+15 & -5-15+20 \\ -2-3+3 & 3+12-9 & 5+15-12 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -2 & 6 & 8 \end{bmatrix}$$

Question No 2

$$\text{Evaluate } \begin{bmatrix} 1 & 2 & 1 \\ 4 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & -4 \\ 1 & 5 \\ -2 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 1 \\ 4 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & -4 \\ 1 & 5 \\ -2 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2+2-2 & -4+10+2 \\ 8+0-4 & -16+0+4 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 8 \\ 4 & -12 \end{bmatrix}$$

$$(ii) \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= \begin{bmatrix} x & y & z \end{bmatrix} \left[ \begin{array}{ccc|c} a & h & g & x \\ h & b & f & y \\ g & f & c & z \end{array} \right]$$

$$= \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} ax + hy + gz \\ hx + by + fz \\ gx + fy + cz \end{bmatrix}$$

$$= \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} ax + hy + gz \\ hx + by + fz \\ gx + fy + cz \end{bmatrix}$$

$$= [ax^2 + hxy + gzx + hxy + by^2 + fyz + gzx + fyz + cz^2]$$

$$= [ax^2 + by^2 + cz^2 + 2hxy + 2fyz + 2gzx]$$

$$(iii) \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}^2$$

$$= \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$$



$$= \begin{bmatrix} 1+8 & 2-6 \\ 4-12 & 8+9 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & -4 \\ -8 & 17 \end{bmatrix}$$

$$(iv) \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}^3$$

$$= \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}^3$$

$$= \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}^2 \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & -4 \\ -8 & 17 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$$

From 2 (iii) above

$$= \begin{bmatrix} 9-16 & 18+12 \\ -8+68 & -16-51 \end{bmatrix}$$

$$= \begin{bmatrix} -7 & 30 \\ 60 & -67 \end{bmatrix}$$

$$(v) \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^2$$

$$= \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1-0-0 & 1-1-0 & 1-0-1 \\ 0+0+0 & 0+1+0 & 0+0+0 \\ 0+0+0 & 0+0+0 & 0+0+1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(vi) \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^3$$

$$= \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



$$\begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^2 \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

by (v) above

$$= \begin{bmatrix} -1+0+0 & -1+0+0 & -1+0+0 \\ 0+0+0 & 0+1+0 & 0+0+0 \\ 0+0+0 & 0+0+0 & 0+0+1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

### Question No 3

Prove that the Product of matrices

$$A = \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix} \text{ and } B = \begin{bmatrix} \cos^2 \phi & \cos \phi \sin \phi \\ \cos \phi \sin \phi & \sin^2 \phi \end{bmatrix}$$

is the zero matrix when  $\theta$  and  $\phi$  differ by an odd multiple of  $\frac{\pi}{2}$ .

$$A = \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix} \text{ and } B = \begin{bmatrix} \cos^2 \phi & \cos \phi \sin \phi \\ \cos \phi \sin \phi & \sin^2 \phi \end{bmatrix}$$

$$AB = \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix} \begin{bmatrix} \cos^2 \phi & \cos \phi \sin \phi \\ \cos \phi \sin \phi & \sin^2 \phi \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2 \theta \cos^2 \phi + \cos \theta \sin \theta \cos \phi \sin \phi & \cos^2 \theta \cos \phi \sin \phi + \cos \theta \sin \theta \sin^2 \phi \\ \cos \theta \sin \theta \cos^2 \phi + \sin^2 \theta \cos \phi \sin \phi & \cos \theta \sin \theta \cos \phi \sin \phi + \sin^2 \theta \sin^2 \phi \end{bmatrix}$$

$$= \begin{bmatrix} \cos \theta \cos \phi (\cos \theta \cos \phi + \sin \theta \sin \phi) & \cos \theta \sin \phi (\cos \theta \cos \phi + \sin \theta \sin \phi) \\ \sin \theta \cos \phi (\cos \theta \cos \phi + \sin \theta \sin \phi) & \sin \theta \sin \phi (\cos \theta \cos \phi + \sin \theta \sin \phi) \end{bmatrix}$$

$$= \begin{bmatrix} \cos \theta \cos \phi \cos(\theta - \phi) & \cos \theta \sin \phi \cos(\theta - \phi) \\ \sin \theta \cos \phi \cos(\theta - \phi) & \sin \theta \sin \phi \cos(\theta - \phi) \end{bmatrix}$$

$$\cos(\theta - \phi) \begin{bmatrix} \cos \theta \cos \phi & \cos \theta \sin \phi \\ \sin \theta \cos \phi & \sin \theta \sin \phi \end{bmatrix} = 0$$

if and only if  $\cos(\theta - \phi) = 0$

Hence

$$AB = 0 \Leftrightarrow \cos(\theta - \phi) = 0$$

$$AB = 0 \Leftrightarrow \theta - \phi = (2n+1) \frac{\pi}{2}$$



### Question NO 5

Show that in general for any two matrices  $A$  and  $B$  which are conformable for addition and multiplication

$$(A+B)^2 \neq A^2 + 2AB + B^2$$

and  $A^2 - B^2 \neq (A-B)(A+B)$

Under what conditions equality holds in each case?

$$(A+B)^2 \neq A^2 + 2AB + B^2$$

$$(A+B)(A+B) = A^2 + AB + BA + B^2$$

Since  $AB \neq BA$  in general, so we must have

$$(A+B)^2 \neq A^2 + 2AB + B^2$$

$$(A-B)(A+B) = A^2 + AB - BA - B^2 \neq A^2 - B^2$$

Since  $AB \neq BA$  is general

Equality holds in each case  $AB = BA$ .

### Question NO 6

If

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}, \text{ show that } A^2 - 4A - 5I =$$

$$\begin{bmatrix} 2 & -2 & 2 \\ 0 & 0 & -2 \\ 0 & 0 & -2 \end{bmatrix} \text{ and } A^3 - 3A^2 - 7A - 3I = 0$$

$$A^2 - 4A - 5I$$

$$= \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - 4 \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+4+2 & 2+2+2 & 1+4+1 \\ 2+2+4 & 4+1+4 & 2+2+2 \\ 2+4+2 & 4+2+2 & 2+4+1 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 4 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 6 & 6 \\ 8 & 9 & 6 \\ 8 & 8 & 7 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 4 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 7-4-5 & 6-8-0 & 6-4-0 \\ 8-8-0 & 9-4-5 & 6-8-0 \\ 8-8-0 & 8-8-0 & 7-4-5 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & -2 & +2 \\ 0 & 0 & -2 \\ 0 & 0 & -2 \end{bmatrix}$$

$$A^3 - 3A^2 - 7A - 3I = A^2 \cdot A - 3A^2 - 7A - 3I$$

$$= \begin{bmatrix} 7 & 6 & 6 \\ 8 & 9 & 6 \\ 8 & 8 & 7 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - 3 \begin{bmatrix} 7 & 6 & 6 \\ 8 & 9 & 6 \\ 8 & 8 & 7 \end{bmatrix} - 7 \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



$$= \begin{bmatrix} 7+16+8 & 6+18+8 & 6+12+7 \\ 14+8+16 & 12+9+16 & 12+6+14 \\ 14+16+8 & 12+18+8 & 12+12+7 \end{bmatrix} = \begin{bmatrix} 21 & 18 & 18 \\ 24 & 27 & 18 \\ 24 & 24 & 21 \end{bmatrix}$$

$$\begin{bmatrix} 7 & 14 & 7 \\ 14 & 7 & 14 \\ 14 & 14 & 7 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 31 & 32 & 25 \\ 38 & 37 & 32 \\ 38 & 38 & 31 \end{bmatrix} = \begin{bmatrix} 21 & 18 & 18 \\ 24 & 27 & 18 \\ 24 & 24 & 21 \end{bmatrix} = \begin{bmatrix} 7 & 14 & 7 \\ 14 & 7 & 14 \\ 14 & 14 & 7 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 31-21-7-3 & 32-18-14-0 & 25-18-7-0 \\ 38-24-14-0 & 37-27-7-3 & 32-27-7-0 \\ 38-24-14-0 & 38-24-14-0 & 31-21-7-3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$$

### Question No 7

Show that the matrix

$$A = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix} \text{ is periodic having Period 2.}$$

$$A^2 = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix} \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 1+6-12 & -2+4+0 & -6-18+18 \\ -3-6+18 & 6+4+0 & 18+18-27 \\ 2+0-6 & -4+0+0 & -12+0+9 \end{bmatrix}$$

$$= \begin{bmatrix} -5 & -6 & -6 \\ 9 & 10 & 9 \\ -4 & -4 & -3 \end{bmatrix}$$

$$A^3 = A^2 \cdot A$$

$$A^3 = \begin{bmatrix} -5 & -6 & -6 \\ 9 & 10 & 9 \\ -4 & -4 & -3 \end{bmatrix} \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} -5+12-12 & 10-12+0 & 30-54-18 \\ 9-30+18 & -18+20+0 & -54+90-27 \\ -4+12-6 & 8-8+0 & 24-36+9 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix} = A$$

Thus A is periodic having Period 2.

Question No 8:

Show that



$\begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}$  is nilpotent. What is nilpotent index?

$$\text{Let } A = \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix} \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} 1+3+4 & -3-9+12 & -4-12+16 \\ -1-3+4 & 3+9-12 & 4+12-16 \\ 1+3-4 & -3-9+12 & -4-12+16 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$$

Hence  $A$  is nilpotent of index 2

Question No 9:

Evaluate

$$\begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix} \text{ is involutory.}$$

Let  $A = \begin{bmatrix} 0 & 1 & 1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$  Then

$$A^2 = \begin{bmatrix} 0 & 1 & 1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 0+4-3 & 0-3+3 & 0+4-4 \\ 0-12+12 & 4+9-12 & -4-12+16 \\ 0-12+12 & 3+9-12 & -3-12+16 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Hence  $A$  is involutory.



### Exercise no 3.2

## Row Operations

- We can interchange any two rows

$$A = \begin{bmatrix} 1 & 2 & -3 & 4 \\ 2 & -1 & -1 & -3 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Interchange  $R_1$  and  $R_3$ ,  $R_1 \leftrightarrow R_3$ ,  $R_{13}$

$$\underline{R_{13}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & -1 & -1 & -3 \\ 1 & 2 & -3 & 4 \end{bmatrix}$$

- We can divide or multiply any Row by a non zero scalar " $k$ ".

- We can add or subtract a scalar multiples of any Row to another row

$$R_2 - 2R_1$$

## Question no 5:

Find the inverse of each of the following matrices

$$(ii) \begin{bmatrix} 1 & 0 & k \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Solution:

$$\begin{bmatrix} 1 & 0 & k \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & k & : & 1 & 0 & 0 \\ 0 & 1 & 0 & : & 0 & 1 & 0 \\ 0 & 0 & 1 & : & 0 & 0 & 1 \end{bmatrix}$$

$$R_1 - kR_3 = \begin{bmatrix} 1 & 0 & 0 & : & 1 & 0 & -k \\ 0 & 1 & 0 & : & 0 & 1 & 0 \\ 0 & 0 & 1 & : & 0 & 0 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1 & 0 & -k \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{Answer}$$

$$\text{iii) } \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 0 & : & 1 & 0 & 0 \\ 0 & 0 & -1 & : & 0 & 1 & 0 \\ 1 & 0 & 0 & : & 0 & 0 & 1 \end{bmatrix}$$

$$R_{13} = \begin{bmatrix} 1 & 0 & 0 & : & 0 & 0 & 1 \\ 0 & 0 & -1 & : & 0 & 1 & 0 \\ 0 & -1 & 0 & : & 1 & 0 & 0 \end{bmatrix}$$

$$R_{23} = \begin{bmatrix} 1 & 0 & 0 & : & 0 & 0 & 1 \\ 0 & -1 & 0 & : & 1 & 0 & 0 \\ 0 & 0 & -1 & : & 0 & 1 & 0 \end{bmatrix}$$

$$-R_2 = \begin{bmatrix} 1 & 0 & 0 & : & 0 & 0 & 1 \\ 0 & 1 & 0 & : & -1 & 0 & 0 \\ 0 & 0 & -1 & : & 0 & 1 & 0 \end{bmatrix}$$



$$-R_3 = \begin{bmatrix} 1 & 0 & 0 & : & 0 & 0 & 1 \\ 0 & 1 & 0 & : & -1 & 0 & 0 \\ 0 & 0 & 1 & : & 0 & -1 & 0 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \text{ Answer.}$$

$$(iii) \begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 2 & -3 & : & 1 & 0 & 0 \\ 2 & 1 & 0 & : & 0 & 1 & 0 \\ 4 & -2 & 5 & : & 0 & 0 & 1 \end{bmatrix}$$

$$-1R_1 = \begin{bmatrix} 1 & -2 & 3 & : & -1 & 0 & 0 \\ 2 & 1 & 0 & : & 0 & 1 & 0 \\ 4 & -2 & 5 & : & 0 & 0 & 1 \end{bmatrix}$$

$$R_2 - 2R_1 = \begin{bmatrix} 1 & -2 & 3 & : & -1 & 0 & 0 \\ 0 & 5 & -6 & : & 2 & 1 & 0 \\ 4 & -2 & 5 & : & 0 & 0 & 1 \end{bmatrix}$$

$$R_3 - 4R_1 = \begin{bmatrix} 1 & -2 & 3 & : & -1 & 0 & 0 \\ 0 & 5 & -6 & : & 2 & 1 & 0 \\ 0 & 6 & 7 & : & 4 & 0 & 1 \end{bmatrix}$$

$$R_3 - R_2 = \begin{bmatrix} 1 & -2 & 3 & : & -1 & 0 & 0 \\ 0 & 5 & -6 & : & 2 & 1 & 0 \\ 0 & 1 & -1 & : & 2 & -1 & 1 \end{bmatrix}$$

$$R_{23} = \begin{bmatrix} 1 & -2 & 3 & : & -1 & 0 & 0 \\ 0 & 1 & -1 & : & 2 & -1 & 1 \\ 0 & 5 & -6 & : & 2 & 1 & 0 \end{bmatrix}$$

$$R_3 - 5R_2 = \begin{bmatrix} 1 & -2 & 3 & : & -1 & 0 & 0 \\ 0 & 1 & -1 & : & 2 & -1 & 1 \\ 0 & 0 & -1 & : & -8 & 6 & -5 \end{bmatrix}$$

$$-1R_3 = \begin{bmatrix} 1 & -2 & 3 & : & -1 & 0 & 0 \\ 0 & 1 & -1 & : & 2 & -1 & 1 \\ 0 & 0 & 1 & : & 8 & -6 & 5 \end{bmatrix}$$

$$R_1 + 2R_2 = \begin{bmatrix} 1 & 0 & 1 & : & 3 & -2 & 2 \\ 0 & 1 & -1 & : & 2 & -1 & 1 \\ 0 & 0 & 1 & : & 8 & -6 & 5 \end{bmatrix}$$

$$R_1 - R_3 = \begin{bmatrix} 1 & 0 & 0 & : & -5 & 4 & -3 \\ 0 & 1 & -1 & : & 2 & -1 & 1 \\ 0 & 0 & 1 & : & 8 & -6 & 5 \end{bmatrix}$$

$$R_2 + R_3 = \begin{bmatrix} 1 & 0 & 0 & : & -5 & 4 & -3 \\ 0 & 1 & 0 & : & 10 & -7 & 6 \\ 0 & 0 & 1 & : & 8 & -6 & 5 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -5 & 4 & -3 \\ 10 & -7 & 6 \\ 8 & -6 & 5 \end{bmatrix} \text{ Answer}$$



$$(iv) \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix}$$

$$\left[ \begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 5 & 2 & -3 & 0 & 0 & 1 \end{array} \right]$$

$$R_3 - 2R_1 = \left[ \begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 1 & 0 & -1 & -2 & 0 & 1 \end{array} \right]$$

$$R_{31} = \left[ \begin{array}{ccc|ccc} 1 & 0 & -1 & 2 & 0 & 1 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 & 0 \end{array} \right]$$

$$R_3 - 2R_1 = \left[ \begin{array}{ccc|ccc} 1 & 0 & -1 & -2 & 0 & 1 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 5 & 0 & 2 \end{array} \right]$$

$$R_2 - R_3 = \left[ \begin{array}{ccc|ccc} 1 & 0 & -1 & -2 & 0 & 1 \\ 0 & 1 & 0 & -5 & 1 & 2 \\ 0 & 1 & 1 & 5 & 0 & -2 \end{array} \right]$$

$$R_3 - R_2 = \left[ \begin{array}{ccc|ccc} 1 & 0 & -1 & -2 & 0 & 1 \\ 0 & 1 & 0 & -5 & 1 & 2 \\ 0 & 0 & 1 & 10 & -1 & -4 \end{array} \right]$$

$$R_1 + R_3 = \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & +8 & -1 & -3 \\ 0 & 1 & 0 & -5 & 1 & 2 \\ 0 & 0 & 1 & 10 & -1 & -4 \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} 8 & -1 & -3 \\ -5 & 1 & 2 \\ 10 & -1 & -4 \end{bmatrix} \text{ Answer.}$$

$$(vii) \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 & : & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & : & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & : & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & : & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_{13} = \begin{bmatrix} 1 & 0 & 0 & 0 & : & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & : & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & : & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & : & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ Answer.}$$



## Echelon and Reduced Echelon form

$$\begin{bmatrix} 1 & 7 & 6 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{bmatrix} \text{ Echelon form}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ Reduced - Echelon}$$

### Echelon Form

An  $m \times n$  matrix  $A$  is said to be Row Echelon form if it satisfied the following conditions.

- all non zero rows are above the any zero row (if exist)
- The number of zero accuring before. The 1st non zero entry. In each non zero is greater then the number of zeros that appear before the 1st non zero Element in any Preceding Row.

$$\begin{bmatrix} 0 & 5 & 2 & 1 \\ 0 & 0 & 9 & -1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

**Zero Row:** If all the elements of a row are zero then the row is said to be Zero Row.

**Pivot or Row Leader:** In an Echelon form the 1st non-zero Entry of a row is called a Pivot Element.

## Reduce Echelon Form

An Echelon form in which each Pivot Element is 1 and every other Entry of Pivot column is zero is said to be reduce Echelon form

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 5 & 0 & 4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

## Exercise no 3.2

### Question NO 6: Reduce Echelon Form

$$\begin{bmatrix} 1 & -2 & 3 & -1 \\ 2 & -1 & 2 & 2 \\ 3 & 1 & 2 & 3 \end{bmatrix}$$

**Solution:**

$$\underline{R_2} \quad \begin{bmatrix} 1 & -2 & 3 & -1 \\ 0 & 3 & -4 & 4 \\ 3 & 1 & 2 & 3 \end{bmatrix} \quad \text{by } R_2 - 2R_1$$



$$\underline{R} \begin{bmatrix} 1 & -2 & 3 & -1 \\ 0 & 3 & -4 & 4 \\ 0 & 7 & -7 & 6 \end{bmatrix} \text{ by } R_3 - 3R_1$$

$$\underline{R} \begin{bmatrix} 1 & -2 & 3 & -1 \\ 0 & 3 & -4 & 4 \\ 0 & 1 & 1 & -2 \end{bmatrix} \text{ by } R_3 - 2R_2$$

$$\underline{R} \begin{bmatrix} 1 & 0 & 5 & -5 \\ 0 & 3 & -4 & 4 \\ 0 & 1 & 1 & -2 \end{bmatrix} \text{ by } R_1 + 2R_3$$

$$\underline{R} \begin{bmatrix} 1 & 0 & 5 & -5 \\ 0 & 0 & -7 & 10 \\ 0 & 1 & 1 & -2 \end{bmatrix} \text{ by } R_2 - 3R_3$$

$$\underline{R} \begin{bmatrix} 1 & 0 & 5 & -5 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & -7 & 10 \end{bmatrix} \text{ by } R_{23}$$

$$\underline{R} \begin{bmatrix} 1 & 0 & 5 & -5 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 1 & -10/7 \end{bmatrix} \text{ by } -\frac{1}{7} R_3$$

$$\underline{R} \begin{bmatrix} 1 & 0 & 0 & 15/7 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -10/7 \end{bmatrix} \text{ by } R_1 - 5R_3$$

$$\underline{R} \begin{bmatrix} 1 & 0 & 0 & 15/7 \\ 0 & 1 & 0 & -4/7 \\ 0 & 0 & 1 & -10/7 \end{bmatrix} \text{ by } R_2 - R_3$$

It is Reduce Echelon form

(ii)

$$\begin{bmatrix} 0 & 1 & 3 & -2 \\ 2 & 1 & -4 & 3 \\ 2 & 3 & 2 & -1 \end{bmatrix}$$

Solution:

$$\xrightarrow{R} \begin{bmatrix} 0 & 1 & 3 & -2 \\ 2 & 1 & -4 & 3 \\ 0 & 2 & 6 & -4 \end{bmatrix} \text{ by } R_3 - R_1$$

$$\xrightarrow{R} \begin{bmatrix} 0 & 1 & 3 & -2 \\ 2 & 0 & -7 & 5 \\ 0 & 2 & 6 & -4 \end{bmatrix} \text{ by } R_2 - R_1$$

$$\xrightarrow{R} \begin{bmatrix} 0 & 1 & 3 & -2 \\ 2 & 0 & -7 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } R_3 - 2R_1$$

$$\xrightarrow{R} \begin{bmatrix} 2 & 0 & -7 & 5 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } R_{12}$$

$$\xrightarrow{R} \begin{bmatrix} 1 & 0 & -7/2 & 5/2 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } \frac{1}{2} R_1$$

It is Reduce Echelon Form.

(iii)

$$\begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 2 & 4 & 1 & -2 & 3 \\ 3 & 6 & 2 & -6 & 3 \end{bmatrix}$$

Solution:



$$R \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 3 & -6 & 1 \\ 3 & 6 & 2 & -6 & 3 \end{bmatrix} \text{ by } R_2 - 2R_1$$

$$R \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 3 & -6 & 1 \\ 0 & 0 & 5 & -12 & 0 \end{bmatrix} \text{ by } R_3 - 3R_1$$

$$R \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 1 & -2 & 1/3 \\ 0 & 0 & 5 & -12 & 0 \end{bmatrix} \text{ by } -\frac{1}{3} R_2$$

$$R \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 1 & -2 & 1/3 \\ 0 & 0 & 1 & -12/5 & 0 \end{bmatrix} \text{ by } \frac{1}{5} R_3$$

$$R \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 5/6 \end{bmatrix} \text{ by } R_2 + 2R_3$$

$$R \begin{bmatrix} 1 & 2 & -1 & 0 & -2/3 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 5/6 \end{bmatrix} \text{ by } R_1 - 2R_3$$

It is Echelon Form

(iv) 
$$\begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 2 & -5 & 3 & 1 \\ 4 & 1 & 1 & 5 \end{bmatrix}$$

Solution:

$$\xrightarrow{R_1} \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 0 & -11 & 5 & -3 \\ 4 & 1 & 1 & 5 \end{bmatrix} \text{ by } R_3 - 2R_1$$

$$R_4 - 4R_1 \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 0 & -11 & 5 & -3 \\ 0 & -11 & 5 & -3 \end{bmatrix}$$

$$\xrightarrow{R_1} \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & -11 & 5 & -3 \end{bmatrix} \text{ by } R_3 + R_2$$

$$\xrightarrow{R_1} \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } R_4 - R_2$$

$$\xrightarrow{R_1} \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 1 & -5/11 & 3/11 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } \frac{1}{11} R_2$$



$$R \begin{bmatrix} 1 & 0 & 4/11 & 13/11 \\ 0 & 1 & -5/11 & 3/11 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } R_1 - 3R_2$$

It is Reduce Echelon Form

Question no 7

Show that

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{R} I_3$$

Solution:

$$R \begin{bmatrix} 1 & 2 & 1 \\ 0 & -5 & -1 \\ 0 & 1 & 2 \end{bmatrix} \text{ by } R_2 - 3R_1$$

$$R \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & -5 & -1 \end{bmatrix} \text{ by } R_{23}$$

$$R \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ 0 & -5 & -1 \end{bmatrix} \text{ by } R_1 - 2R_2$$

$$R \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 9 \end{bmatrix} \text{ by } R_3 + 5R_2$$

$$\underline{B} \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \text{ by } \frac{1}{9} R_1$$

$$\underline{R} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \text{ by } R_1 + 3R_3$$

$$\underline{R} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ by } R_2 - 2R_3$$

$$= I_3$$

Question no 8:

Find the rank of each of the following matrices

$$i. \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ 5 & -1 \\ -2 & 3 \end{bmatrix}$$

Solution:

$$\underline{R} \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ 0 & -16 \\ -2 & 3 \end{bmatrix} \text{ by } R_3 - 5R_1$$

$$\underline{R} \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ 0 & -16 \\ 0 & 9 \end{bmatrix} \text{ by } R_4 + 2R_1$$



$$R \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ 0 & -16 \\ 0 & 9 \end{bmatrix} \text{ by } -\frac{1}{2} R_2$$

$$R \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ 0 & 1 \\ 0 & 9 \end{bmatrix} \text{ by } -\frac{1}{16} R_3$$

$$R \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \text{ by } \frac{1}{9} R_4$$

$$R \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \text{ by } R_3 - R_2$$

$$R \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \text{ by } R_1 - R_2$$

The rank of the given matrix is 2

(ii)  $\begin{bmatrix} 1 & 2 & -3 \\ 2 & 1 & 0 \\ -2 & -1 & 3 \\ -1 & 4 & -2 \end{bmatrix}$

Solution:

$$R \begin{bmatrix} 1 & 2 & -3 \\ 0 & -3 & 6 \\ -2 & -1 & 3 \\ -1 & 4 & -2 \end{bmatrix} \text{ by } R_2 - 2R_1$$

$$\underline{R} \begin{bmatrix} 1 & 2 & -3 \\ 0 & -3 & 6 \\ 0 & 3 & -3 \\ -1 & 4 & -2 \end{bmatrix} \quad \text{by } R_3 + 2R_1$$

$$\underline{R} \begin{bmatrix} 1 & 2 & -3 \\ 0 & -3 & 6 \\ 0 & 3 & -3 \\ 0 & 6 & -5 \end{bmatrix} \quad \text{by } R_4 + R_1$$

$$\underline{R} \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & 2 \\ 0 & 3 & -3 \\ 0 & 6 & -5 \end{bmatrix} \quad \text{by } -\frac{1}{3} R_2$$

$$\underline{R} \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & 2 \\ 0 & 1 & -1 \\ 0 & 6 & -5 \end{bmatrix} \quad \text{by } \frac{1}{3} R_3$$

$$\underline{R} \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & -3 \\ 0 & 6 & -5 \end{bmatrix} \quad \text{by } R_3 - R_2$$

$$\underline{R} \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & -3 \\ 0 & 0 & -17 \end{bmatrix} \quad \text{by } R_4 - 6R_2$$



$$\underline{R} \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & -17 \end{bmatrix} \text{ by } -\frac{1}{3} R_3$$

$$\underline{R} \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \text{ by } -\frac{1}{17} R_4$$

$$\underline{R} \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \text{ by } R_4 - R_3$$

The Rank of this matrix is 1

(iii)

$$\begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 1 & 4 & 3 & -1 & -4 \\ 2 & 3 & -4 & -7 & -3 \\ 3 & 8 & 1 & -7 & -8 \end{bmatrix}$$

Solution.

$$\underline{R} \begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 0 & 1 & 2 & 1 & -1 \\ 2 & 3 & -4 & -7 & -3 \\ 3 & 8 & 1 & -7 & -8 \end{bmatrix} \text{ by } R_2 - R_1$$

$$\underline{R} \begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 0 & 1 & 2 & 1 & -1 \\ 0 & -3 & -6 & -3 & 3 \\ 3 & 8 & 1 & -7 & -8 \end{bmatrix} \text{ by } R_3 - 2R_1$$

$$\underline{R} \begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 0 & 1 & 2 & 1 & -1 \\ 0 & -3 & -6 & -3 & 3 \\ 0 & -1 & -2 & -1 & 1 \end{bmatrix} \text{ by } R_4 - 3R_1$$

$$\underline{R} \begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 0 & 1 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & -2 & -1 & 1 \end{bmatrix} \text{ by } R_3 + R_2$$

$$\underline{R} \begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 0 & 1 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } R_4 - R_2$$

The rank of the given matrix is 2

(iv)

$$\begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 1 & 4 & 3 & 1 & -4 \\ 2 & 3 & -4 & -7 & -3 \\ 3 & 8 & 1 & -7 & -8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & -2 & 5 & 4 \\ 1 & 4 & 1 & 3 & 5 \\ 1 & 4 & 2 & 4 & 3 \\ 2 & 7 & -3 & 6 & 13 \end{bmatrix}$$

Solution:



$$\underline{R} \begin{bmatrix} 1 & 3 & -2 & 5 & 4 \\ 0 & 1 & 3 & -2 & 1 \\ 1 & 4 & 2 & 4 & 3 \\ 2 & 7 & -3 & 6 & 13 \end{bmatrix} \text{ by } R_3 - R_1$$

$$\underline{R} \begin{bmatrix} 1 & 3 & -2 & 5 & 4 \\ 0 & 1 & 3 & -2 & 1 \\ 0 & 1 & 4 & -1 & -1 \\ 2 & 7 & -3 & 6 & 13 \end{bmatrix} \text{ by } R_3 - R_1$$

$$\underline{R} \begin{bmatrix} 1 & 3 & -2 & 5 & 4 \\ 0 & 1 & 3 & -2 & 1 \\ 0 & 1 & 4 & -1 & -1 \\ 0 & 1 & 1 & -4 & 5 \end{bmatrix} \text{ by } R_4 - 2R_1$$

$$\underline{R} \begin{bmatrix} 1 & 3 & -2 & 5 & 4 \\ 0 & 1 & 3 & -2 & 1 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 1 & 1 & -4 & 5 \end{bmatrix} \text{ by } R_3 - R_2$$

$$\underline{R} \begin{bmatrix} 1 & 3 & -2 & 5 & 4 \\ 0 & 1 & 3 & -2 & 1 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 2 & -2 & 4 \end{bmatrix} \text{ by } R_4 - R_2$$

$$\underline{R} \begin{bmatrix} 1 & 3 & -2 & 5 & 4 \\ 0 & 1 & 3 & -2 & 1 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } R_4 + 2R_3$$

The rank of this matrix is 3.

# System of Equation

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

## Example 4.1.

Question: Show that gauss Elimination Method.

$$x_1 - x_2 + 2x_3 = 0$$

$$4x_1 + x_2 + 2x_3 = 01$$

$$x_1 + x_2 + x_3 = -1$$

Solution:

The augmented form

$$Ab = \left[ \begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 4 & 1 & 2 & 1 \\ 1 & 1 & 1 & -1 \end{array} \right]$$

$$\xrightarrow{R} \left[ \begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 0 & 5 & -6 & 1 \\ 0 & 2 & -1 & -1 \end{array} \right] \begin{array}{l} \text{by } R_2 - 4R_1 \\ R_3 - R_1 \end{array}$$



$$B \left[ \begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 0 & 1 & -4 & 3 \\ 0 & 2 & -1 & -1 \end{array} \right] \text{ by } R_3 - 2R_2$$

$$R \left[ \begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 0 & 1 & -4 & 3 \\ 0 & 0 & 7 & -1 \end{array} \right] \text{ by } R_3 - 2R_2$$

$$R \left[ \begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 0 & 1 & -4 & 3 \\ 0 & 0 & 1 & -1 \end{array} \right] \text{ by } \frac{1}{7} R_3$$

$$x_1 - x_2 + 2x_3 = 0 \quad \text{--- (i)}$$

$$0x_1 + x_2 - 4x_3 = 3 \quad \text{--- (ii)}$$

$$0x_1 + 0x_2 + x_3 = -1 \quad \text{--- (iii)}$$

from eq (iii)

$$\boxed{x_3 = -1}$$

$x_3 = -1$ , Put in eq (ii)

$$x_2 - 4(-1) = 3$$

$$x_2 + 4 = 3$$

$$x_2 = 3 - 4$$

$$\boxed{x_2 = -1}$$

Put  $x_2 = -1$ ,  $x_3 = -1$  in eq (i)

$$x_1 + 1 - 2 = 0$$

$$x_1 = 2 - 1$$

$$x_1 = 1$$

$$x_1 = 1, \quad x_2 = -1, \quad x_3 = -1$$

Question: Solve system of equation by Gauss-Elimination method.

$$x_1 - x_2 + 2x_3 = 0$$

$$4x_1 + x_2 + 2x_3 = 1$$

$$x_1 + x_2 + x_3 = -1$$

Solution:

The augmented form

$$\text{Ab} \begin{bmatrix} 1 & -1 & 2 & : & 0 \\ 4 & 1 & 2 & : & 1 \\ 1 & 1 & 1 & : & -1 \end{bmatrix}$$

$$\underline{R} \begin{bmatrix} 1 & -1 & 2 & : & 0 \\ 0 & 5 & -6 & : & 1 \\ 0 & 2 & -1 & : & -1 \end{bmatrix} \begin{array}{l} \text{by } R_2 - 4R_1 \\ R_3 - R_1 \end{array}$$

$$\underline{R} \begin{bmatrix} 1 & -1 & 2 & : & 0 \\ 0 & 1 & -4 & : & 3 \\ 0 & 2 & -1 & : & -1 \end{bmatrix} \text{ by } R_2 - 2R_3$$

$$\underline{R} \begin{bmatrix} 1 & -1 & 2 & : & 0 \\ 0 & 1 & -4 & : & 3 \\ 0 & 0 & 7 & : & 7 \end{bmatrix} \text{ by } R_3 - 2R_2$$



$$R \left[ \begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 0 & 1 & -4 & 3 \\ 0 & 0 & 1 & -1 \end{array} \right] \text{ by } \frac{1}{-1} R_2$$

$$R \left[ \begin{array}{ccc|c} 1 & 0 & -2 & 3 \\ 0 & 1 & -4 & 3 \\ 0 & 0 & 1 & -1 \end{array} \right] R_1 + R_2$$

$$R \left[ \begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array} \right] \text{ by } R_1 + 2R_3, R_2 + R_3$$

$$x_1 = 1, x_2 = -1, x_3 = -1$$

Question: Solve system of equation by Gauss-Elimination Method.

$$x_1 - x_2 + 2x_3 + x_4 = 0$$

$$3x_1 + 2x_3 + x_4 = 0$$

$$4x_1 + x_2 + 2x_3 + 2x_4 = 0$$

Solution:

The Augmented form

$$Ab = \left[ \begin{array}{cccc|c} 1 & -1 & 2 & 1 & 0 \\ 3 & 2 & 0 & 1 & 0 \\ 4 & 1 & 2 & 2 & 0 \end{array} \right]$$

$$\underline{R} \left[ \begin{array}{cccc|c} 1 & -1 & 2 & 1 & 0 \\ 0 & 5 & -6 & -2 & 0 \\ 0 & 5 & -6 & -2 & 0 \end{array} \right] \begin{array}{l} \text{by } R_2 - 3R_1 \\ R_3 - 4R_1 \end{array}$$

$$\underline{R} \left[ \begin{array}{cccc|c} 1 & -1 & 2 & 1 & 0 \\ 0 & 5 & -6 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \text{by } R_3 - R_2$$

Solution is infinite

### Guass - Jordan

$$\underline{R} \left[ \begin{array}{cccc|c} 1 & -1 & 2 & 1 & 0 \\ 0 & 1 & -6/5 & -2/5 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \text{by } \frac{1}{5} R_2$$

$$\underline{R} \left[ \begin{array}{cccc|c} 1 & 0 & 4/5 & 3/5 & 0 \\ 0 & 1 & -6/5 & -2/5 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$x_1 + \frac{4}{5}x_3 + \frac{3}{5}x_4 = 0 \quad \text{--- (i)}$$

$$x_2 - \frac{6}{5}x_3 - \frac{2}{5}x_4 = 0 \quad \text{--- (ii)}$$



$$x_1 = -\frac{4}{5}x_3 - \frac{3}{5}x_4$$

$$x_2 = \frac{6}{5}x_3 + \frac{2}{5}x_4$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -4/5 x_3 - 3/5 x_4 \\ 6/5 x_3 + 2/5 x_4 \\ x_3 \\ x_4 \end{bmatrix}$$

$$= \begin{bmatrix} -4/5 x_3 \\ 6/5 x_3 \\ x_3 \\ 0 \end{bmatrix} + \begin{bmatrix} -3/5 x_4 \\ 2/5 x_4 \\ 0 \\ x_4 \end{bmatrix}$$

$$= \begin{bmatrix} -4/5 \\ 6/5 \\ 1 \\ 0 \end{bmatrix} x_3 + \begin{bmatrix} -3/5 \\ 2/5 \\ 0 \\ 1 \end{bmatrix} x_4$$

$\xrightarrow{a} \quad \quad \quad \xrightarrow{b}$

$$x_3 = a$$

$x_4 = b$  for arbitrary value of  
 $a$  &  $b$

## LU-Decomposition / LU Factorization

Consider

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$AX = B$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \rightarrow \begin{matrix} \text{lower triangle} \\ \text{matrix} \end{matrix}$$

$$U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \rightarrow \begin{matrix} \text{upper triangle} \\ \text{matrix} \end{matrix}$$

Step #1

$$LU = A$$

Step #2

$$UX = Y$$

Step #3:

$$LY = B$$



## Group

Let  $(G, *)$  be a non-empty set is said to group if it satisfied following axioms.

(i) Closure Law

(ii) Associative Law

(iii) Identity Law

(iv) Inverse Law

(v) Commutative Law

Show that  $G = \{1, -1\}$  is an abelian group under ' $\times$ '

(i) Closure Law

$$\forall a, b \in G \text{ s.t. } a \times b \in G$$

$$\text{Let } (1, -1) \in G$$

$$1 \times (-1) = -1 \in G$$

Closure Law hold w.r.t. ( $\times$ )

(ii) Associative Law

$$\forall a, b, c \in G \text{ s.t.}$$

$$a \times (b \times c) = (a \times b) \times c$$

$$\text{Let } 1, -1, -1$$

$$1 \times (-1 \times -1) = (1 \times -1) \times -1$$

$$1 \times 1 = -1 \times -1$$

$$1 = 1$$

Associative Law hold

### (iii) Identity law

$$\forall a \in G \exists a' \in G \text{ s.t.}$$

$$axa' = a'xa = e$$

$$\text{Let } 1 \in G = 1 \in G$$

$$1 \times 1 = 1$$

$$\text{Let } -1 \in G = -1 \in G$$

$$-1 \times -1 = 1$$

### (iv) Commutative law

$$\forall a, b \in G$$

$$axb = bxa$$

$$\text{Let } -1, 1$$

$$-1 \times 1 = 1 \times -1$$

$$-1 = -1$$

### (v) Inverse law

$$\forall a \in G \exists a^{-1} \in G$$

$$axa^{-1} = a^{-1}xa = a$$

$$\text{Let}$$

$$-1 \in G, 1 \in G$$

$$-1 \times 1 = 1 \times -1$$

$$-1 = -1$$



## Vector Space

Let  $V$  be a non-empty set and  $F$  be a field

Suppose for every  $a, b \in F$  and  $u, v \in V$  Then

$V$  is called vector space over  $F$ . If it satisfied following axioms.

1.  $V$  is an abelian group under "+"

2.  $a + v \in V$

$a = \text{Scalar}$

3.  $a(u+v) = au + av$

$v = \text{Vector}$

4.  $(a+b)u = au + bu$

5.  $a(bu) = (ab)u$

6.  $1 \cdot v = v \cdot 1 = v$

Question: Show that  $\mathbb{R}^2$  is a vector space over  $\mathbb{R}$

Solution:

$$\mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}$$

1.  $(\mathbb{R}^2, +)$  is an abelian group  $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$

(i) Closure Law

Let  $u, v \in \mathbb{R}^2$  s.t

$$u + v \in \mathbb{R}^2$$

$$U = (x_1, y_1)$$

$$V = (x_2, y_2)$$

$$U+V = (x_1, y_1) + (x_2, y_2)$$

$$U+V = (x_1+x_2, y_1+y_2) \in \mathbb{R}^2$$

Hold this property

### ii) Associative Law

$$\text{Let } U, V, W \in \mathbb{R}^2$$

$$U+(V+W) = (U+V)+W$$

L.H.S

$$U+(V+W) = (x_1, y_1) + ((x_2, y_2) + (x_3, y_3))$$

$$= (x_1, y_1) + (x_2+x_3, y_2+y_3)$$

$$= (x_1+x_2+x_3, y_1+y_2+y_3)$$

$$= (x_1+x_2, y_1+y_2) + (x_3, y_3)$$

$$= (U+V)+W = \text{R.H.S}$$

Hold this property

### (iii) Identity Law

$$\forall U \in \mathbb{R}^2 \exists e \in \mathbb{R}^2$$

$$U+e = e+U = U$$

$$e = (0, 0), U = (x_1, y_1)$$

$$U+e = (x_1, y_1) + (0, 0)$$



$$= ((x_1 + 0), (y_1 + 0))$$

$$= (x_1 + 0, y_1 + 0)$$

$$= (x_1, y_1)$$

$$u + e = u$$

Hold this property

#### (iv) Inverse law

$$u \in \mathbb{R}^2 \quad \exists \quad u' \in \mathbb{R}^2$$

$$u + u' = e$$

$$\text{Let } u = (x_1, y_1) \in \mathbb{R}^2, -u = (-x_1, -y_1)$$

$$u + (-u) = (x_1, y_1) + (-x_1, -y_1)$$

$$= (x_1 - x_1, y_1 - y_1)$$

$$u + (-u) = (0, 0)$$

Holds this property

#### (v) Commutative law

$$\forall u, v \in \mathbb{R}^2$$

$$u + v = v + u$$

L.H.S

$$u + v = (x_1, y_1) + (x_2, y_2)$$

$$= (x_1 + x_2, y_1 + y_2)$$

$$= (x_2 + x_1, y_2 + y_1)$$

$$v + u = \text{R.H.S}$$

All axioms of abelian group are satisfied

$\mathbb{R}^2$  is an abelian group

Axiom (1) is True

(2)  $\forall u \in \mathbb{R}^2$

$\forall u \in \mathbb{R}^2$  and  $a \in F$

$$v = (x, y)$$

$$au = a(x, y)$$

$$av = (ax, ay) \in \mathbb{R}^2$$

$$a(u+v) = au + av$$

$$a \in F, u, v \in \mathbb{R}^2$$

$$u = (x_1, y_1), v = (x_2, y_2)$$

L.H.S

$$a(u+v) = a[(x_1, y_1) + (x_2, y_2)]$$

$$= a(x_1 + x_2, y_1 + y_2)$$

$$a(u+v) = (ax_1 + ax_2, ay_1 + ay_2)$$

$$(ax_1, ay_1) + (ax_2, ay_2)$$

$$a(x_1, y_1) + a(x_2, y_2)$$

$$a(u) + a(v) \text{ R.H.S}$$

Axiom (3) is true.



2.  $a \in \mathbb{R}^2$

$\forall u \in \mathbb{R}^2$  and  $a \in F$

$$v = (x, y)$$

$$au = a(x_1, y_1)$$

$$av = (ax_1, ay_1) \in \mathbb{R}^2$$

$$a(u+v) = au + av$$

$$a \in F, u, v \in \mathbb{R}^2$$

$$u = (x_1, y_1), v = (x_2, y_2)$$

L.H.S

$$a(u+v) = a[(x_1, y_1) + (x_2, y_2)]$$

$$= a(x_1 + x_2, y_1 + y_2)$$

$$a(u+v) = (ax_1 + ax_2, ay_1 + ay_2)$$

$$= (ax_1, ay_1) + (ax_2, ay_2)$$

$$a(x_1, y_1) + a(x_2, y_2)$$

$$a(u) + a(v) \quad \text{R.H.S}$$

Axiom (3) is true

$$(a+b)u = au + bu$$

$$\forall a, b \in \mathbb{R}, \forall u \in \mathbb{R}^2$$

$$u = (x_1, y_1)$$

$$(a+b)u = (a+b)(x_1, y_1)$$

$$((a+b)x_1, (a+b)y_1)$$

$$(a+b)u = ax_1 + bx_1, ay_1 + by_1$$

$$(ax_1, ay_1) + (bx_1, by_1)$$

$$a(x_1, y_1) + b(x_1, y_1)$$

$$au + bu \text{ R.H.S}$$

Axiom 4 true.

$$(5) a(bu) = ab(u)$$

$$\forall a, b \in R, u \in R^2$$

$$u = (x_1, y_1)$$

$$a(bu) = a(b(x_1, y_1))$$

$$a(bx_1, by_1)$$

$$abx_1, aby_1$$

$$(abx_1, aby_1)$$

$$ab(x_1, y_1)$$

$$a(bu) = ab(u) \text{ R.H.S}$$

Axiom (5) is true

$$(6) 1 \cdot v = v \cdot 1 = u$$

$$1 \in R, v \in R^2$$

$$\text{Let } v = (x_1, y_1)$$

$$\text{L.H.S}$$



$$1 \cdot v = 1(x_1, y_1)$$

$$= (x_1, y_1)$$

$$= v$$

R.H.S

Axiom (6) is true

Hence all axioms of vector space are satisfied

So  $\mathbb{R}^2$  is a vector space over  $\mathbb{R}$ .